

Q1. Let $\lambda, \mu \in \mathbb{R}$. If the system of equations
 $3x + 5y + \lambda z = 3$
 $7x + 11y - 9z = 2$
 $97x + 155y - 189z = \mu$
 has infinitely many solutions, then $\mu + 2\lambda$ is equal to : 09 Apr 2024 (M)

- (1) 24 (2) 25
 (3) 22 (4) 27

Q2. Consider the matrices : $A = \begin{bmatrix} 2 & -5 \\ 3 & m \end{bmatrix}$, $B = \begin{bmatrix} 20 \\ m \end{bmatrix}$ and $X = \begin{bmatrix} x \\ y \end{bmatrix}$. Let the set of all m , for which the system of equations $AX = B$ has a negative solution (i.e., $x < 0$ and $y < 0$), be the interval (a, b) . Then $8 \int_a^b |A| dm$ is equal to 09 Apr 2024 (E)

Q3. If $\alpha \neq a, \beta \neq b, \gamma \neq c$ and $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$, then $\frac{a}{\alpha-a} + \frac{b}{\beta-b} + \frac{\gamma}{\gamma-c}$ is equal to: 08 Apr 2024 (E)

- (1) 3 (2) 0
 (3) 1 (4) 2

Q4. If the system of equations $x + 4y - z = \lambda, 7x + 9y + \mu z = -3, 5x + y + 2z = -1$ has infinitely many solutions, then $(2\mu + 3\lambda)$ is equal to : 08 Apr 2024 (E)

- (1) 3 (2) -3
 (3) -2 (4) 2

Q5. If the system of equations
 $2x + 7y + \lambda z = 3$
 $3x + 2y + 5z = 4$
 $x + \mu y + 32z = -1$
 has infinitely many solutions, then $(\lambda - \mu)$ is equal to 06 Apr 2024 (E)

Q6. Let $\alpha\beta\gamma = 45; \alpha, \beta, \gamma \in \mathbb{R}$. If $x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ for some $x, y, z \in \mathbb{R}, xyz \neq 0$, then $6\alpha + 4\beta + \gamma$ is equal to 06 Apr 2024 (M)

Q7. For $\alpha, \beta \in \mathbb{R}$ and a natural number n , let $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r - 2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$. Then 06 Apr 2024 (M)

- (1) 0 (2) $4\alpha + 2\beta$
 (3) $2\alpha + 4\beta$ (4) $2n$

Q8. If the system of equations
 $11x + y + \lambda z = -5$
 $2x + 3y + 5z = 3$
 $8x - 19y - 39z = \mu$
 has infinitely many solutions, then $\lambda^4 - \mu$ is equal to : 05 Apr 2024 (M)

- (1) 51 (2) 45
 (3) 47 (4) 49

Q9. The values of m, n , for which the system of equations

$$x + y + z = 4,$$

$$2x + 5y + 5z = 17,$$

$$x + 2y + mz = n$$

has infinitely many solutions, satisfy the equation:

05 Apr 2024 (E)

(1) $m^2 + n^2 - mn = 39$

(2) $m^2 + n^2 - m - n = 46$

(3) $m^2 + n^2 + m + n = 64$

(4) $m^2 + n^2 + mn = 68$

Q10. If the system of equations

$$x + (\sqrt{2} \sin \alpha)y + (\sqrt{2} \cos \alpha)z = 0$$

$$x + (\cos \alpha)y + (\sin \alpha)z = 0$$

$$x + (\sin \alpha)y - (\cos \alpha)z = 0$$

has a non-trivial solution, then $\alpha \in (0, \frac{\pi}{2})$ is equal to :

04 Apr 2024 (M)

(1) $\frac{11\pi}{24}$

(2) $\frac{5\pi}{24}$

(3) $\frac{7\pi}{24}$

(4) $\frac{3\pi}{4}$

Q11. Let the system of equations $x + 2y + 3z = 5, 2x + 3y + z = 9, 4x + 3y + \lambda z = \mu$ have infinite number of solutions. Then $\lambda + 2\mu$ is equal to:

01 Feb 2024 (E)

(1) 28

(2) 17

(3) 22

(4) 15

Q12. If the system of equations

$$2x + 3y - z = 5$$

$$x + \alpha y + 3z = -4$$

$$3x - y + \beta z = 7$$

has infinitely many solutions, then $13\alpha\beta$ is equal to

01 Feb 2024 (M)

(1) 1110

(2) 1120

(3) 1210

(4) 1220

Q13.

If $f(x) = \begin{vmatrix} x^3 & 2x^2 + 1 & 1 + 3x \\ 3x^2 + 2 & 2x & x^3 + 6 \\ x^3 - x & 4 & x^2 - 2 \end{vmatrix}$ for all $x \in \mathbb{R}$, then $2f(0) + f'(0)$ is equal to

31 Jan 2024 (M)

(1) 48

(2) 24

(3) 42

(4) 18

Q14. If the system of linear equations

$$x - 2y + z = -4$$

$$2x + \alpha y + 3z = 5$$

$$3x - y + \beta z = 3$$

has infinitely many solutions, then $12\alpha + 13\beta$ is equal to

31 Jan 2024 (M)

(1) 60

(3) 54

(2) 64

(4) 58

Q15.

Let A be a 3×3 real matrix such that $A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $A \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = 4 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. Then, the

system $(A - 3I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ has

31 Jan 2024 (E)

(1) unique solution

(2) exactly two solutions

(3) no solution

(4) infinitely many solutions

Q16.

If $f(x) = \begin{vmatrix} 2 \cos^4 x & 2 \sin^4 x & 3 + \sin^2 2x \\ 3 + 2 \cos^4 x & 2 \sin^4 x & \sin^2 2x \\ 2 \cos^4 x & 3 + 2 \sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5} f'(0)$ is equal to _____.

30 Jan 2024 (M)

(1) 0

(2) 1

(3) 2

(4) 6

Q17. Consider the system of linear equation $x + y + z = 4\mu$, $x + 2y + 2\lambda z = 10\mu$, $x + 3y + 4\lambda^2 z = \mu^2 + 15$,

where $\lambda, \mu \in \mathbb{R}$. Which one of the following statements is NOT correct?

30 Jan 2024 (M)

(1) The system has unique solution if $\lambda \neq \frac{1}{2}$ and $\mu \neq 1, 15$

(2) The system is inconsistent if $\lambda = \frac{1}{2}$ and $\mu \neq 1$

(3) The system has infinite number of solutions if $\lambda = \frac{1}{2}$ and $\mu = 15$

(4) The system is consistent if $\lambda \neq \frac{1}{2}$

Q18. Consider the system of linear equations $x + y + z = 5$, $x + 2y + \lambda^2 z = 9$ and $x + 3y + \lambda z = \mu$, where

$\lambda, \mu \in \mathbb{R}$. Then, which of the following statement is NOT correct?

30 Jan 2024 (E)

(1) System has infinite number of solution if $\lambda = 1$ and $\mu = 13$

(2) System is inconsistent if $\lambda = 1$ and $\mu \neq 13$

(3) System has unique solution if $\lambda \neq 1$ and $\mu \neq 13$

(4) System is consistent if $\lambda \neq 1$ and $\mu = 13$

Q19. Let for any three distinct consecutive terms a, b, c of an A.P, the lines $ax + by + c = 0$ be concurrent at the point P and $Q(\alpha, \beta)$ be a point such that the system of equations $x + y + z = 6$, $2x + 5y + \alpha z = \beta$ and $x + 2y + 3z = 4$, has infinitely many solutions. Then $(PQ)^2$ is equal to _____.

29 Jan 2024 (E)

Q20.

The values of α , for which $\begin{vmatrix} 1 & \frac{3}{2} & \alpha + \frac{3}{2} \\ 1 & \frac{1}{3} & \alpha + \frac{1}{3} \\ 2\alpha + 3 & 3\alpha + 1 & 0 \end{vmatrix} = 0$, lie in the interval

27 Jan 2024 (E)

(1) $(-2, 1)$

(2) $(-3, 0)$

(3) $(-\frac{3}{2}, \frac{3}{2})$

(4) $(0, 3)$

Q21. Let the system of linear equations

$$-x + 2y - 9z = 7$$

$$-x + 3y + 7z = 9$$

$$-2x + y + 5z = 8$$

$$-3x + y + 13z = \lambda$$

has a unique solution $x = \alpha$, $y = \beta$, $z = \gamma$. Then the distance of the point (α, β, γ) from the plane $2x - 2y + z = \lambda$ is

15 Apr 2023 (M)

(1) 11

(3) 9

(2) 7

(4) 13

Q22. For the system of linear equations

$$2x + 4y + 2az = b$$

$$x + 2y + 3z = 4$$

$$2x + 5y + 2z = 8$$

which of the following is NOT correct?

13 Apr 2023 (M)

(1) It has unique solution if $a = b = 6$

(2) It has infinitely many solutions if $a = 3$, $b = 6$

(3) It has infinitely many solutions if $a = 3$, $b = 8$

(4) It has unique solution if $a = b = 8$

Q23. If the system of equations

$$2x + y - z = 5$$

$$2x - 5y + \lambda z = \mu$$

$$x + 2y - 5z = 7$$

has infinitely many solutions, then $(\lambda + \mu)^2 + (\lambda - \mu)^2$ is equal to

13 Apr 2023 (E)

(1) 904

(3) 912

(2) 916

(4) 920

Q24.

Let $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$. If $\sum_{k=1}^n D_k = 96$, then n is equal to _____.

12 Apr 2023 (M)

Q25.

If $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$, then λ , $\frac{\lambda}{3}$ are the roots of the equation

11 Apr 2023 (E)

(1) $4x^2 + 24x - 27 = 0$

(2) $4x^2 - 24x - 27 = 0$

(3) $4x^2 + 24x + 27 = 0$

(4) $4x^2 - 24x + 27 = 0$

Q26. If the system of linear equations

$$7x + 11y + \alpha z = 13$$

$$5x + 4y + 7z = \beta$$

$$175x + 194y + 57z = 361$$

has infinitely many solutions, then $\alpha + \beta + 2$ is equal to

11 Apr 2023 (E)

(1) 4

(2) 3

(3) 5

(4) 6

Q27. For the system of linear equations

$$2x - y + 3z = 5$$

$$3x + 2y - z = 7$$

$$4x + 5y + \alpha z = \beta,$$

which of the following is NOT correct?

10 Apr 2023 (M)

(1) The system has infinitely many solutions for $\alpha = -5$ and $\beta = 9$

(2) The system has infinitely many solutions for $\alpha = -6$ and $\beta = 9$

(3) The system is inconsistent for $\alpha = -5$ and $\beta = 8$

(4) The system has a unique solution for $\alpha \neq -5$ and $\beta = 8$

Q28. Let S be the set of values of λ , for which the system of equations

$6\lambda x - 3y + 3z = 4\lambda^2$, $2x + 6\lambda y + 4z = 1$ and $3x + 2y + 3\lambda z = \lambda$ has no solution. Then, $12 \sum_{\lambda \in S} |\lambda|$ is equal to _____.

10 Apr 2023 (E)

Q29. Let S be the set of all values of $\theta \in [-\pi, \pi]$ for which the system of linear equations

$$x + y + \sqrt{3}z = 0$$

$$-x + (\tan \theta)y + \sqrt{7}z = 0$$

$$x + y + (\tan \theta)z = 0$$

has non-trivial solution. Then $\frac{120}{\pi} \sum_{\theta \in S} \theta$ is equal to

08 Apr 2023 (E)

(1) 20

(2) 40

(3) 30

(4) 10

Q30. If the system of equations

$$x + y + az = b$$

$$2x + 5y + 2z = 6$$

$$x + 2y + 3z = 3$$

has infinitely many solutions, then $2a + 3b$ is equal to

06 Apr 2023 (M)

(1) 25

(2) 20

(3) 23

(4) 28

Q31. For the system of equations

$$x + y + z = 6$$

$$x + 2y + \alpha z = 10$$

$x + 3y + 5z = \beta$, which one of the following is NOT true?

06 Apr 2023 (E)

- (1) System has no solution for $\alpha = 3, \beta = 24$
 (2) System has a unique solution for $\alpha = -3, \beta = 14$
 (3) System has infinitely many solutions for $\alpha = 3, \beta = 14$
 (4) System has a unique solution for $\alpha = 3, \beta \neq 14$

Q32. Let $f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$, $x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$. If α and β respectively are the maximum and the minimum values of f , then

01 Feb 2023 (M)

- (1) $\beta^2 - 2\sqrt{\alpha} = \frac{19}{4}$ (2) $\beta^2 + 2\sqrt{\alpha} = \frac{19}{4}$
 (3) $\alpha^2 - \beta^2 = 4\sqrt{3}$ (4) $\alpha^2 + \beta^2 = \frac{9}{2}$

Q33. Let S denote the set of all real values of λ such that the system of equations

$$\lambda x + y + z = 1$$

$$x + \lambda y + z = 1$$

$$x + y + \lambda z = 1$$

is inconsistent, then $\sum_{\lambda \in S} (|\lambda|^2 + |\lambda|)$ is equal to

01 Feb 2023 (M)

- (1) 2 (2) 12
 (3) 4 (4) 6

Q34. For the system of linear equations $ax + y + z = 1, x + ay + z = 1, x + y + az = \beta$, which one of the following statements is NOT correct?

01 Feb 2023 (E)

- (1) It has infinitely many solutions if $\alpha = 2$ and $\beta = -1$
 (2) It has no solution if $\alpha = -2$ and $\beta = 1$
 (3) $x + y + z = \frac{3}{4}$ if $\alpha = 2$ and $\beta = 1$
 (4) It has infinitely many solutions if $\alpha = 1$ and $\beta = 1$

Q35. If a point $P(\alpha, \beta, \gamma)$ satisfying $(\alpha \ \beta \ \gamma) \begin{pmatrix} 2 & 10 & 8 \\ 9 & 3 & 8 \\ 8 & 4 & 8 \end{pmatrix} = (0 \ 0 \ 0)$ lies on the plane $2x + 4y + 3z = 5$, then

31 Jan 2023 (E)

$6\alpha + 9\beta + 7\gamma$ is equal to

- (1) $\frac{5}{4}$ (2) -1
 (3) 11 (4) $\frac{11}{5}$

Q36. For the system of linear equations

$$x + y + z = 6$$

$$\alpha x + \beta y + 7z = 3$$

$$x + 2y + 3z = 14$$

which of the following is NOT true?

31 Jan 2023 (M)

- (1) If $\alpha = \beta = 7$, then the system has no solution
- (2) If $\alpha = \beta$ and $\alpha \neq 7$ then the system has a unique solution.
- (3) There is a unique point (α, β) on the line $x + 2y + 18 = 0$ for which the system has infinitely many solutions
- (4) For every point $(\alpha, \beta) \neq (7, 7)$ on the line $x - 2y + 7 = 0$, the system has infinitely many solutions.

Q37. For $\alpha, \beta \in \mathbb{R}$, suppose the system of linear equations

$$x - y + z = 5$$

$$2x + 2y + \alpha z = 8$$

$$3x - y + 4z = \beta$$

has infinitely many solutions. Then α and β are the roots of

$$(1) x^2 - 10x + 16 = 0$$

$$(2) x^2 + 18x + 56 = 0$$

$$(3) x^2 - 18x + 56 = 0$$

$$(4) x^2 + 14x + 24 = 0$$

30 Jan 2023 (E)

Q38. Let the system of linear equations

$$x + y + kz = 2$$

$$2x + 3y - z = 1$$

$$3x + 4y + 2z = k$$

have infinitely many solutions. Then the system

$$(k+1)x + (2k-1)y = 7$$

$$(2k+1)x + |k+5|y = 10 \text{ has :}$$

(1) infinitely many solutions

(2) unique solution satisfying $x - y = 1$

(3) no solution

(4) unique solution satisfying $x + y = 1$

30 Jan 2023 (M)

Q39. Consider the following system of questions

$$\alpha x + 2y + z = 1$$

$$2\alpha x + 3y + z = 1$$

$$3x + \alpha y + 2z = \beta$$

For some $\alpha, \beta \in \mathbb{R}$. Then which of the following is NOT correct.

(1) It has no solution if $\alpha = -1$ and $\beta \neq 2$

(2) It has no solution for $\alpha = -1$ and for all $\beta \in \mathbb{R}$

(3) It has no solution for $\alpha = 3$ and for all $\beta \neq 2$

(4) It has a solution for all $\alpha \neq -1$ and $\beta = 2$

29 Jan 2023 (M)

Q40. Let S_1 and S_2 be respectively the sets of all $a \in \mathbb{R} - \{0\}$ for which the system of linear equations

$$ax + 2ay - 3az = 1$$

$$(2a+1)x + (2a+3)y + (a+1)z = 2$$

$$(3a+5)x + (a+5)y + (a+2)z = 3$$

has unique solution and infinitely many solutions. Then

(1) $n(S_1) = 2$ and S_2 is an infinite set

(2) S_1 is an infinite set and $n(S_2) = 2$

(3) $S_1 = \phi$ and $S_2 = \mathbb{R} - \{0\}$

(4) $S_1 = \mathbb{R} - \{0\}$ and $S_2 = \phi$

25 Jan 2023 (M)

Q41. If the system of equations

$$x + 2y + 3z = 3,$$

$$4x + 3y - 4z = 4 \text{ and}$$

$$8x + 4y - \lambda z = 9 + \mu \text{ has infinitely many solutions, then the ordered pair } (\lambda, \mu) \text{ is equal to}$$

24 Jan 2023 (E)

(1) $\left(\frac{72}{5}, \frac{21}{5}\right)$

(2) $\left(\frac{-72}{5}, \frac{-21}{5}\right)$

(3) $\left(\frac{72}{5}, \frac{-21}{5}\right)$

(4) $\left(\frac{-72}{5}, \frac{21}{5}\right)$

Q42. Let α be a root of the equation $(a - c)x^2 + (b - a)x + (c - b) = 0$ where a, b, c are distinct real numbers

such that the matrix $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ is singular. Then the value of $\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$ is

24 Jan 2023 (M)

(1) 6

(2) 3

(3) 9

(4) 12

Q43. Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then

29 Jul 2022 (M)

(1) The system of linear equations $AX = 0$ has a unique solution

(2) The system of linear equations $AX = 0$ has infinitely many solutions

(3) B is an invertible matrix

(4) $\text{adj}(A)$ is an invertible matrix

Q44. If the system of equations

$$x + y + z = 6$$

$$2x + 5y + \alpha z = \beta$$

$$x + 2y + 3z = 14$$

has infinitely many solutions, then $\alpha + \beta$ is equal to

29 Jul 2022 (E)

(1) 8

(2) 36

(3) 44

(4) 48

Q45. Let p and $p + 2$ be prime numbers and let

$$\Delta = \begin{vmatrix} p! & (p+1)! & (p+2)! \\ (p+1)! & (p+2)! & (p+3)! \\ (p+2)! & (p+3)! & (p+4)! \end{vmatrix}$$

Then the sum of the maximum values of α and β , such that p^α and $(p+2)^\beta$ divide Δ , is _____.

29 Jul 2022 (M)

Q46. If the system of linear equations.

$$8x + y + 4z = -2$$

$$x + y + z = 0$$

$$\lambda x - 3y = \mu$$

has infinitely many solutions, then the distance of the point $(\lambda, \mu, -\frac{1}{2})$ from the plane $8x + y + 4z + 2 = 0$ is:

26 Jul 2022 (M)

(1) $3\sqrt{5}$

(3) $\frac{26}{9}$

(2) 4

(4) $\frac{10}{3}$

Q47. The number of $\theta \in (0, 4\pi)$ for which the system of linear equations

$$3(\sin 3\theta)x - y + z = 2$$

$$3(\cos 2\theta)x + 4y + 3z = 3$$

$$6x + 7y + 7z = 9$$
 has no solution is

(1) 6

(2) 7

(3) 8

(4) 9

25 Jul 2022 (M)

Q48. The number of real values of λ , such that the system of linear equations

$$2x - 3y + 5z = 9$$

$$x + 3y - z = -18$$

$$3x - y + (\lambda^2 - |\lambda|)z = 16$$
 has no solutions, is

(1) 0

(2) 1

(3) 2

(4) 4

25 Jul 2022 (E)

Q49. If the system of linear equations

$$2x + y - z = 7$$

$$x - 3y + 2z = 1$$

$$x + 4y + \delta z = k$$
, where $\delta, k \in \mathbb{R}$

has infinitely many solutions, then $\delta + k$ is equal to

(1) -3

(2) 3

(3) 6

(4) 9

29 Jun 2022 (M)

Q50. If the system of linear equations

$$2x + 3y - z = -2$$

$$x + y + z = 4$$

$$x - y + |\lambda|z = 4\lambda - 4$$
 where $\lambda \in \mathbb{R}$,

has no solution, then

(1) $\lambda = 7$

(2) $\lambda = -7$

(3) $\lambda = 8$

(4) $\lambda^2 = 1$

28 Jun 2022 (M)

Q51. If the system of linear equations

$$2x - 3y = \gamma + 5$$

$\alpha x + 5y = \beta + 1$, where $\alpha, \beta, \gamma \in \mathbb{R}$ has infinitely many solutions, then the value of $|9\alpha + 3\beta + 5\gamma|$ is equal to

28 Jun 2022 (E)

Q52. Let the system of linear equations $x + 2y + z = 2$, $\alpha x + 3y - z = \alpha$, $-\alpha x + y + 2z = -\alpha$ be inconsistent.

Then α is equal to

27 Jun 2022 (M)

(1) $\frac{5}{2}$
(3) $\frac{7}{2}$

(2) $-\frac{5}{2}$
(4) $-\frac{7}{2}$

Q53.

Let $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a \in R$. Then the sum of the squares of all the values of a for

$2f'(10) - f'(5) + 100 = 0$ is

27 Jun 2022 (E)

(1) 117
(3) 125

(2) 106
(4) 136

Q54. If the system of equations $\alpha x + y + z = 5$, $x + 2y + 3z = 4$, $x + 3y + 5z = \beta$. Has infinitely many solutions, then the ordered pair (α, β) is equal to

26 Jun 2022 (E)

(1) (1, -3)
(3) (1, 3)

(2) (-1, 3)
(4) (-1, -3)

Q55. The ordered pair (a, b) , for which the system of linear equations

$3x - 2y + z = b$
 $5x - 8y + 9z = 3$
 $2x + y + az = -1$
has no solution, is

26 Jun 2022 (M)

(1) $(3, \frac{1}{3})$
(3) $(-3, -\frac{1}{3})$

(2) $(-3, \frac{1}{3})$
(4) $(3, -\frac{1}{3})$

Q56. The system of equations

$-kx + 3y - 14z = 25$
 $-15x + 4y - kz = 3$
 $-4x + y + 3z = 4$
is consistent for all k in the set

25 Jun 2022 (E)

(1) R
(3) $R - \{-13\}$

(2) $R - \{-11, 13\}$
(4) $R - \{-11, 11\}$

Q57.

Let A be a 3×3 real matrix such that $A \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$; $A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$ and $A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$. If

$X = [x_1 \ x_2 \ x_3]^T$ and I is an identity matrix of order 3, then the system $(A - 2I)X = \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}$ has

25 Jun 2022 (M)

(1) no solution
(3) unique solution

(2) infinitely many solutions
(4) exactly two solutions

Q58. The number of values of α for which the system of equations

$x + y + z = \alpha$

$$\alpha x + 2\alpha y + 3z = -1$$

$$x + 3\alpha y + 5z = 4$$

is inconsistent, is

24 Jun 2022 (M)

(1) 0

(3) 2

(2) 1

(4) 3

Q59. Let the system of linear equations

$$x + y + \alpha z = 2$$

$$3x + y + z = 4$$

$$x + 2z = 1$$

have a unique solution (x^*, y^*, z^*) . If (α, x^*) , (y^*, α) and $(x^*, -y^*)$ are collinear points, then the sum of absolute values of all possible values of α is

24 Jun 2022 (E)

(1) 4

(3) 2

(2) 3

(4) 1

Q60. Consider the system of linear equations

$$-x + y + 2z = 0$$

$$3x - ay + 5z = 1$$

$$2x - 2y - az = 7$$

Let S_1 be the set of all $a \in \mathbb{R}$ for which the system is inconsistent and S_2 be the set of all $a \in \mathbb{R}$ for which the system has infinitely many solutions. If $n(S_1)$ and $n(S_2)$ denote the number of elements in S_1 and S_2 respectively, then

01 Sep 2021 (E)

(1) $n(S_1) = 2, n(S_2) = 0$

(3) $n(S_1) = 0, n(S_2) = 2$

(2) $n(S_1) = 2, n(S_2) = 2$

(4) $n(S_1) = 1, n(S_2) = 0$

Q61. If the following system of linear equations

$$2x + y + z = 5$$

$$x - y + z = 3$$

$$x + y + az = b$$

has no solution, then :

31 Aug 2021 (M)

(1) $a = -\frac{1}{3}, b \neq \frac{7}{3}$

(3) $a \neq -\frac{1}{3}, b = \frac{7}{3}$

(2) $a \neq \frac{1}{3}, b = \frac{7}{3}$

(4) $a = \frac{1}{3}, b \neq \frac{7}{3}$

Q62. If $\alpha + \beta + \gamma = 2\pi$, then the system of equations

$$x + (\cos \gamma)y + (\cos \beta)z = 0$$

$$(\cos \gamma)x + y + (\cos \alpha)z = 0$$

$$(\cos \beta)x + (\cos \alpha)y + z = 0$$

has :

31 Aug 2021 (E)

(1) infinitely many solutions

(3) no solution

(2) a unique solution

(4) exactly two solutions

Q63. If $a_r = \cos \frac{2r\pi}{9} + i \sin \frac{2r\pi}{9}$, $r = 1, 2, 3, \dots, i = \sqrt{-1}$, then the determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is equal to : 31 Aug 2021 (M)

(1) a_9

(2) $a_1a_9 - a_3a_7$

(3) a_5

(4) $a_2a_6 - a_4a_8$

Q64. If the system of linear equations
 $2x + y - z = 3$
 $x - y - z = \alpha$
 $3x + 3y + \beta z = 3$
 has infinitely many solutions, then $|\alpha + \beta - \alpha\beta|$ is equal to _____. 27 Aug 2021 (M)

Q65. Let $[\lambda]$ be the greatest integer less than or equal to λ . The set of all values of λ for which the system of linear equations $x + y + z = 4$, $3x + 2y + 5z = 3$, $9x + 4y + (28 + [\lambda])z = [\lambda]$ has a solution is: 27 Aug 2021 (E)

(1) R

(2) $(-\infty, -9) \cup [-8, \infty)$

(3) $(-\infty, -9) \cup (-9, \infty)$

(4) $[-9, -8)$

Q66. Let $A = \begin{bmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{bmatrix}$, where $[x]$ denotes the greatest integer less than or equal to x . If $\det(A) = 192$, then the set of values of x is in the interval: 27 Aug 2021 (E)

(1) $[62, 63)$

(2) $[65, 66)$

(3) $[60, 61)$

(4) $[68, 69)$

Q67. Let $\theta \in (0, \frac{\pi}{2})$. If the system of linear equations
 $(1 + \cos^2 \theta)x + \sin^2 \theta y + 4 \sin 3\theta z = 0$,
 $\cos^2 \theta x + (1 + \sin^2 \theta)y + 4 \sin 3\theta z = 0$,
 $\cos^2 \theta x + \sin^2 \theta y + (1 + 4 \sin 3\theta)z = 0$
 has a non-trivial solution, then the value of θ is: 26 Aug 2021 (M)

(1) $\frac{4\pi}{9}$

(2) $\frac{5\pi}{18}$

(3) $\frac{7\pi}{18}$

(4) $\frac{\pi}{18}$

Q68. Let $f(x) = \begin{vmatrix} \sin^2 x & -2 + \cos^2 x & \cos 2x \\ 2 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1 + \cos 2x \end{vmatrix}$, $x \in [0, \pi]$. Then the maximum value of $f(x)$ is equal to 27 Jul 2021 (M)

Q69. For real numbers α and β , consider the following system of linear equations: $x + y - z = 2$, $x + 2y + \alpha z = 1$ and $2x - y + z = \beta$. If the system has infinite solutions, then $\alpha + \beta$ is equal to _____. 27 Jul 2021 (M)

Q70. The values of a and b , for which the system of equations

$2x + 3y + 6z = 8$

$$x + 2y + az = 5$$

$$3x + 5y + 9z = b$$

has no solution, are :

25 Jul 2021 (M)

(1) $a = 3, b \neq 13$

(2) $a \neq 3, b \neq 13$

(3) $a \neq 3, b = 3$

(4) $a = 3, b = 13$

Q71. The values of λ and μ such that the system of equations $x + y + z = 6$, $3x + 5y + 5z = 26$ and $x + 2y + \lambda z = \mu$ has no solution, are:

22 Jul 2021 (M)

(1) $\lambda = 3, \mu = 5$

(2) $\lambda = 3, \mu \neq 10$

(3) $\lambda \neq 2, \mu = 10$

(4) $\lambda = 2, \mu \neq 10$

Q72.

Let a, b, c, d be in arithmetic progression with common difference λ . If $\begin{vmatrix} x+a-c & x+b & x+a \\ x-1 & x+c & x+b \\ x-b+d & x+d & x+c \end{vmatrix} = 2$, then value of λ^2 is equal to _____.

20 Jul 2021 (M)

Q73. The value of $k \in R$, for which the following system of linear equations

$$3x - y + 4z = 3$$

$$x + 2y - 3z = -2$$

$$6x + 5y + kz = -3$$

has infinitely many solutions, is:

20 Jul 2021 (E)

(1) 3

(2) -5

(3) 5

(4) -3

Q74.

Let $A = [a_{ij}]$ be a 3×3 matrix, where $a_{ij} = \begin{cases} 1 & \text{if } i = j \\ -x & \text{if } |i - j| = 1 \\ 2x + 1 & \text{otherwise} \end{cases}$

Let a function $f : R \rightarrow R$ be defined as $f(x) = \det(A)$. Then the sum of maximum and minimum values of f on R is equal to:

20 Jul 2021 (M)

(1) $-\frac{20}{27}$

(2) $\frac{88}{27}$

(3) $\frac{20}{27}$

(4) $-\frac{88}{27}$

Q75. Let α, β, γ be the real roots of the equation, $x^3 + ax^2 + bx + c = 0$, ($a, b, c \in R$ and $a, b \neq 0$). If the system of equations (in u, v, w) given by

$$\alpha u + \beta v + \gamma w = 0,$$

$$\beta u + \gamma v + \alpha w = 0,$$

$$\gamma u + \alpha v + \beta w = 0$$

has non-trivial solution, then the value of $\frac{a^2}{b}$ is

18 Mar 2021 (M)

(1) 5

(2) 3

(3) 1

(4) 0

Q76. Let the system of linear equations

$$4x + \lambda y + 2z = 0$$

$$2x - y + z = 0$$

$$\mu x + 2y + 3z = 0, \lambda, \mu \in R$$

has a non-trivial solution. Then which of the following is true?

(1) $\mu = 6, \lambda \in R$

(2) $\lambda = 2, \mu \in R$

(3) $\lambda = 3, \mu \in R$

(4) $\mu = -6, \lambda \in R$

18 Mar 2021 (E)

Q77. The system of equations $kx + y + z = 1$, $x + ky + z = k$ and $x + y + zk = k^2$ has no solution if k is equal to:

(1) 0

(2) 1

(3) -1

(4) -2

17 Mar 2021 (M)

Q78. If x, y, z are in arithmetic progression with common difference d , $x \neq 3d$, and the determinant of the matrix

$$\begin{bmatrix} 3 & 4\sqrt{2} & x \\ 4 & 5\sqrt{2} & y \\ 5 & k & z \end{bmatrix}$$
 is zero, then the value of k^2 is

(1) 72

(2) 12

(3) 36

(4) 6

17 Mar 2021 (E)

Q79. Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$, $i = \sqrt{-1}$. Then, the system of linear equations $A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has :

(1) A unique solution

(2) Infinitely many solutions

(3) No solution

(4) Exactly two solutions

16 Mar 2021 (M)

Q80.

The maximum value of $f(x) = \begin{vmatrix} \sin^2 x & 1 + \cos^2 x & \cos 2x \\ 1 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & \sin 2x \end{vmatrix}$, $x \in R$ is

(1) $\sqrt{7}$

(2) $\frac{3}{4}$

(3) $\sqrt{5}$

(4) 5

16 Mar 2021 (E)

Q81. Consider the following system of equations:

$$x + 2y - 3z = a$$

$$2x + 6y - 11z = b$$

$$x - 2y + 7z = c$$

where a, b and c are real constants. Then the system of equations :

(1) has a unique solution when $5a = 2b + c$

(2) has no solution for all a, b and c

(3) has infinite number of solutions when

(4) has a unique solution for all a, b and c

$$5a = 2b + c$$

26 Feb 2021 (E)

Q82.

The value of $\begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix}$ is

(1) 0

(2) $(a+2)(a+3)(a+4)$

(3) -2

(4) $(a+1)(a+2)(a+3)$

26 Feb 2021 (M)

Q83. If the system of equations

$$kx + y + 2z = 1$$

$$3x - y - 2z = 2$$

$$-2x - 2y - 4z = 3$$

has infinitely many solutions, then k is equal to _____.

25 Feb 2021 (M)

Q84. The following system of linear equations

$$2x + 3y + 2z = 9$$

$$3x + 2y + 2z = 9$$

$$x - y + 4z = 8$$

(1) has infinitely many solutions

(2) has a unique solution

(3) has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^3 = 12$

(4) does not have any solution

25 Feb 2021 (E)

Q85. Let A be a 3×3 matrix with $\det(A) = 4$. Let R_i denote the i^{th} row of A . If a matrix B is obtained by performing the operation $R_2 \rightarrow 2R_2 + 5R_3$ on A , then $\det(B)$ is equal to :

25 Feb 2021 (E)

(1) 64

(2) 16

(3) 128

(4) 80

Q86. The system of linear equations

$$3x - 2y - kz = 10$$

$$2x - 4y - 2z = 6$$

$$x + 2y - z = 5m$$

is inconsistent if :

24 Feb 2021 (M)

(1) $k = 3, m \neq \frac{4}{5}$

(2) $k = 3, m = \frac{4}{5}$

(3) $k \neq 3, m \in R$

(4) $k \neq 3, m \neq \frac{4}{5}$

Q87. For the system of linear equations:

$$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in R$$

Consider the following statements:

(A) The system has unique solution if $k \neq 2, k \neq -2$.

(B) The system has unique solution if $k = -2$.

(C) The system has unique solution if $k = 2$.

(D) The system has no-solution if $k = 2$.

(E) The system has infinite number of solutions if $k \neq -2$.

Which of the following statements are correct?

24 Feb 2021 (E)

(1) (A) and (E) only

(2) (B) and (E) only

(3) (A) and (D) only

(4) (C) and (D) only

Q88. Let A and B be 3×3 real matrices such that A is a symmetric matrix and B is a skew-symmetric matrix. Then the system of linear equations $(A^2 B^2 - B^2 A^2)X = O$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has 24 Feb 2021 (E)

- (1) exactly two solutions (2) infinitely many solutions
(3) a unique solution (4) no solution

Q89. The values of λ and μ for which the system of linear equations $x + y + z = 2$, $x + 2y + 3z = 5$, $x + 3y + \lambda z = \mu$ has infinitely many solutions, are respectively 06 Sep 2020 (M)

- (1) 6 and 8 (2) 5 and 7
(3) 5 and 8 (4) 4 and 9

Q90. Let m and M be respectively the minimum and maximum value values of $\begin{vmatrix} \cos^2 x & 1 + \sin^2 x & \sin 2x \\ 1 + \cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1 + \sin 2x \end{vmatrix}$. Then the ordered pair (m, M) is equal to: 06 Sep 2020 (M)

- (1) (3, 3) (2) (-3, -1)
(3) (4, 1) (4) (1, 3)

Q91. The sum of distinct values of λ for which the system of equations :

$$\begin{aligned} (\lambda - 1)x + (3\lambda + 1)y + 2\lambda z &= 0 \\ (\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z &= 0 \\ 2x + (3\lambda + 1)y + 3(\lambda - 1)z &= 0, \end{aligned}$$

Has non-zero solutions, is

06 Sep 2020 (E)

Q92. If $a + x = b + y = c + z + 1$, where a, b, c, x, y, z are non-zero distinct real numbers, then

$\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix}$ is equal to : 05 Sep 2020 (E)

(1) $y(b-a)$ (2) $y(a-b)$
(3) 0 (4) $y(a-c)$

Q93. If the minimum and the maximum values of the function $f : \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$, defined by

$f(\theta) = \begin{vmatrix} -\sin^2 \theta & -1 - \sin^2 \theta & 1 \\ -\cos^2 \theta & -1 - \cos^2 \theta & 1 \\ 12 & 10 & -2 \end{vmatrix}$ are m and M respectively, then the ordered pair (m, M) is equal to : 05 Sep 2020 (M)

- (1) $(0, 2\sqrt{2})$ (2) $(-4, 0)$
(3) $(-4, 4)$ (4) $(0, 4)$

Q94. Let $\lambda \in \mathbb{R}$. The system of linear equations

$$\begin{aligned} 2x_1 - 4x_2 + \lambda x_3 &= 1 \\ x_1 - 6x_2 + x_3 &= 2 \end{aligned}$$

$$\lambda x_1 - 10x_2 + 4x_3 = 3$$

is inconsistent for :

05 Sep 2020 (M)

(1) exactly one positive value of λ

(2) exactly one negative value of λ

(3) every value of λ

(4) exactly two values of λ

Q95. If the system of linear equations

$$x + y + 3z = 0$$

$$x + 3y + k^2z = 0$$

$$3x + y + 3z = 0$$

has a non-zero solution (x, y, z) for some $k \in \mathbb{R}$, then $x + \left(\frac{y}{z}\right)$ is equal to :

05 Sep 2020 (E)

(1) -3

(2) 9

(3) 3

(4) -9

Q96. If the system of equations

$$x + y + z = 2$$

$$2x + 4y - z = 6$$

$$3x + 2y + \lambda z = \mu$$

has infinitely many solutions, then :

04 Sep 2020 (E)

(1) $\lambda + 2\mu = 14$

(2) $2\lambda - \mu = 5$

(3) $\lambda - 2\mu = -5$

(4) $2\lambda + \mu = 14$

Q97. If the system of equations

$$x - 2y + 3z = 9$$

$$2x + y + z = b$$

$x - 7y + az = 24$, has infinitely many solutions, then $a - b$ is equal to _____

04 Sep 2020 (M)

Q98. Suppose the vectors x_1, x_2 and x_3 are the solutions of the system of linear equations, $Ax = b$ when the vector

b on the right side is equal to b_1, b_2 and b_3 respectively. If $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$;

$b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$, $b_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$, then the determinant of A is equal to

04 Sep 2020 (E)

(1) 4

(2) 2

(3) $\frac{1}{2}$

(4) $\frac{3}{2}$

Q99.

If $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$, then $B + C$ is equal to :

03 Sep 2020 (M)

(1) -1

(2) 1

(3) -3

(4) 9

Q100. Let S be the set of all integer solutions (x, y, z) of the system of equations

$$x - 2y + 5z = 0$$

$$-2x + 4y + z = 0$$

$$-7x + 14y + 9z = 0$$

such that $15 \leq x^2 + y^2 + z^2 \leq 150$. Then, the number of elements in the set S is equal to

03 Sep 2020 (E)

Q101. Let S be the set of all $\lambda \in R$ for which the system of linear equations

$$2x - y + 2z = 2$$

$$x - 2y + \lambda z = -4$$

$$x + \lambda y + z = 4$$

has no solution. Then the set S

02 Sep 2020 (M)

(1) Contains more than two elements

(2) Is an empty set

(3) Is a singleton

(4) Contains exactly two elements

Q102. Let $a - 2b + c = 1$.

$$\text{If } f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}, \text{ then:}$$

09 Jan 2020 (E)

$$(1) f(-50) = 501$$

$$(2) f(-50) = -1$$

$$(3) f(50) = -501$$

$$(4) f(50) = 1$$

Q103. If for some α and β in R , the intersection of the following three planes

$$x + 4y - 2z = 1$$

$$x + 7y - 5z = \beta$$

$$x + 5y + \alpha z = 5$$

is a line in R^3 , then $\alpha + \beta$ is equal to:

09 Jan 2020 (M)

$$(1) 0$$

$$(2) 10$$

$$(3) 2$$

$$(4) -10$$

Q104. The following system of linear equations

$$7x + 6y - 2z = 0$$

$$3x + 4y + 2z = 0$$

$$x - 2y - 6z = 0, \text{ has}$$

09 Jan 2020 (E)

(1) infinitely many solutions, (x, y, z) satisfying $y = 2z$

(2) no solution

(3) infinitely many solutions, (x, y, z) satisfying $x = 2z$

(4) only the trivial solution

Q105. For which of the following ordered pairs (μ, δ) , the system of linear equations

$$x + 2y + 3z = 1$$

$$3x + 4y + 5z = \mu$$

$$4x + 4y + 4z = \delta$$

is inconsistent?

08 Jan 2020 (M)

(1) (4, 3)

(2) (4, 6)

(3) (1, 0)

(4) (3, 4)

Q106. The system of linear equations

$$\lambda x + 2y + 2z = 5$$

$$2\lambda x + 3y + 5z = 8$$

$$4x + \lambda y + 6z = 10$$

08 Jan 2020 (E)

(1) no solution when $\lambda = 8$

(2) a unique solution when $\lambda = -8$

(3) no solution when $\lambda = 2$

(4) infinitely many solutions when $\lambda = 2$

Q107. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)} a_{ij}$, where $i, j = 1, 2, 3$. If the determinant of B is 81, then determinant of A is

07 Jan 2020 (E)

(1) $\frac{1}{3}$

(2) 3

(3) $\frac{1}{81}$

(4) $\frac{1}{9}$

Q108. If the system of linear equations,

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$3x + 2y + \lambda z = \mu$$

has more than two solutions, then $\mu - \lambda^2$, is equal to.

07 Jan 2020 (E)

Q109. If the system of linear equations

$$2x + 2ay + az = 0$$

$$2x + 3by + bz = 0$$

$$2x + 4cy + cz = 0,$$

where $a, b, c \in \mathbb{R}$ are non-zero and distinct; has a non-zero solution, then

07 Jan 2020 (M)

(1) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A. P.

(2) a, b, c are in G. P.

(3) $a + b + c = 0$

(4) a, b, c are in A. P.

Q110.

A value of $\theta \in (0, \frac{\pi}{3})$, for which $\begin{vmatrix} 1 + \cos^2\theta & \sin^2\theta & 4 \cos 6\theta \\ \cos^2\theta & 1 + \sin^2\theta & 4 \cos 6\theta \\ \cos^2\theta & \sin^2\theta & 1 + 4 \cos 6\theta \end{vmatrix} = 0$, is

12 Apr 2019 (E)

(1) $\frac{\pi}{9}$

(2) $\frac{7\pi}{24}$

(3) $\frac{7\pi}{36}$

(4) $\frac{\pi}{18}$

Q111. If $[x]$ denotes the greatest integer $\leq x$, then the system of linear equations $[\sin\theta]x + [-\cos\theta]y = 0$,

$$[\cot\theta]x + y = 0$$

12 Apr 2019 (E)

(1) has a unique solution if $\theta \in (\frac{\pi}{2}, \frac{2\pi}{3}) \cup (\pi, \frac{7\pi}{6})$

(2) have infinitely many solution if $\theta \in (\frac{\pi}{2}, \frac{2\pi}{3}) \cup (\pi, \frac{7\pi}{6})$

(3) has a unique if $\theta \in (\frac{\pi}{2}, \frac{2\pi}{3})$ and have infinitely many solutions if $\theta \in (\pi, \frac{7\pi}{6})$

(4) have infinitely many solutions if $\theta \in (\frac{\pi}{2}, \frac{2\pi}{3})$ and has a unique solution if $\theta \in (\pi, \frac{7\pi}{6})$

Q112. If the system of linear equations $x + y + z = 5$, $x + 2y + 2z = 6$, $x + 3y + \lambda z = \mu$, ($\lambda, \mu \in \mathbb{R}$), has infinitely many solutions, then the value of $\lambda + \mu$ is:

10 Apr 2019 (M)

- (1) 7 (2) 10
(3) 12 (4) 9

Q113. The sum of the real roots of the equation

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0, \text{ is equal to:}$$

10 Apr 2019 (E)

- (1) 0 (2) -4
(3) 6 (4) 1

Q114. Let λ be a real number for which the system of linear equations

$$\begin{aligned} x + y + z &= 6, \\ 4x + \lambda y - \lambda z &= \lambda - 2 \text{ and} \\ 3x + 2y - 4z &= -5 \end{aligned}$$

has infinitely many solutions. Then λ is a root of the quadratic equation:

10 Apr 2019 (E)

- (1) $\lambda^2 + 3\lambda - 4 = 0$ (2) $\lambda^2 - \lambda - 6 = 0$
(3) $\lambda^2 - 3\lambda - 4 = 0$ (4) $\lambda^2 + \lambda - 6 = 0$

Q115. If $\Delta_1 = \begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}$, $x \neq 0$; then for all $\theta \in (0, \frac{\pi}{2})$:

10 Apr 2019 (M)

- (1) $\Delta_1 + \Delta_2 = -2(x^3 + x - 1)$ (2) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$
(3) $\Delta_1 + \Delta_2 = -2x^3$ (4) $\Delta_1 - \Delta_2 = -2x^3$

Q116.

Let α and β be the roots of the equation $x^2 + x + 1 = 0$. Then for $y \neq 0$ in $\mathbb{R}$, $\begin{vmatrix} y+1 & \alpha & \beta \\ \alpha & y+\beta & 1 \\ \beta & 1 & y+\alpha \end{vmatrix}$ is

09 Apr 2019 (M)

equal to

- (1) y^3 (2) $y(y^2 - 1)$
(3) $y^3 - 1$ (4) $y(y^2 - 3)$

Q117. If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution

(x, y, z) , then $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to

09 Apr 2019 (E)

- (1) $-\frac{1}{4}$ (2) $\frac{1}{2}$
(3) -4 (4) $\frac{3}{4}$

Q118. The greatest value of $c \in \mathbb{R}$ for which the system of linear equations $x - cy - cz = 0$, $cx - y + cz = 0$,

$cx + cy - z = 0$ has a non-trivial solution, is

08 Apr 2019 (M)

- (1) -1 (2) 2
(3) $\frac{1}{2}$ (4) 0

Q119. Let the numbers 2, b , c be in an A.P. and $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}$. If $\det(A) \in [2, 16]$, then c lies in the interval:

08 Apr 2019 (E)

- (1) $[2, 3]$
 (2) $[4, 6]$
 (3) $\left[3, 2 + 2^{\frac{3}{4}}\right]$
 (4) $\left(2 + 2^{\frac{3}{4}}, 4\right)$

Q120. If the system of linear equations

$$x - 2y + kz = 1$$

$$2x + y + z = 2$$

$$3x - y - kz = 3$$

has a solution (x, y, z) , $z \neq 0$, then (x, y) lies on the straight line whose equation is:

08 Apr 2019 (E)

- (1) $4x - 3y - 4 = 0$
 (2) $3x - 4y - 4 = 0$
 (3) $3x - 4y - 1 = 0$
 (4) $4x - 3y - 1 = 0$

Q121. If $A = \begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{bmatrix}$, then for all $\theta \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$, $\det(A)$ lies in the interval :

12 Jan 2019 (E)

- (1) $\left(1, \frac{5}{2}\right]$
 (2) $\left[\frac{5}{2}, 4\right)$
 (3) $\left(\frac{3}{2}, 3\right]$
 (4) $\left(0, \frac{3}{2}\right]$

Q122. An ordered pair (α, β) for which the system of linear equations

$$(1 + \alpha)x + \beta y + z = 2$$

$$\alpha x + (1 + \beta)y + z = 3$$

$\alpha x + \beta y + 2z = 2$ has a unique solution, is :

12 Jan 2019 (M)

- (1) $(-3, 1)$
 (2) $(1, -3)$
 (3) $(2, 4)$
 (4) $(-4, 2)$

Q123. The set of all values of λ for which the system of linear equations

$$x - 2y - 2z = \lambda x$$

$$x + 2y + z = \lambda y$$

$$-x - y = \lambda z$$

has a non-trivial solution :

12 Jan 2019 (E)

- (1) is an empty set
 (2) contains more than two elements
 (3) is a singleton
 (4) contains exactly two elements

Q124. If the system of linear equations $2x + 2y + 3z = a$

$$3x - y + 5z = b$$

$$x - 3y + 2z = c$$

where, a, b, c are non-zero real numbers, has more than one solution, then

11 Jan 2019 (M)

- (1) $b - c + a = 0$
 (2) $b - c - a = 0$
 (3) $a + b + c = 0$
 (4) $b + c - a = 0$

Q125. If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)(x+a+b+c)^2$, $x \neq 0$ and $a+b+c \neq 0$, then x

is equal to

11 Jan 2019 (E)

- (1) abc (2) $-(a+b+c)$
(3) $2(a+b+c)$ (4) $-2(a+b+c)$

Q126. Let A and B be two invertible matrices of order 3×3 . If $\det(ABA^T) = 8$ and $\det(AB^{-1}) = 8$, then $\det(BA^{-1}B^T)$ is

equal to

11 Jan 2019 (E)

- (1) $\frac{1}{4}$ (2) 1
(3) $\frac{1}{16}$ (4) 16

Q127. Let $d \in \mathbb{R}$, and $A = \begin{bmatrix} -2 & 4+d & (\sin \theta) - 2 \\ 1 & (\sin \theta) + 2 & d \\ 5 & (2 \sin \theta) - d & (-\sin \theta) + 2 + 2d \end{bmatrix}$, $\theta \in [0, 2\pi]$. If the minimum value of $\det(A)$ is 8, then a value of d is:

10 Jan 2019 (M)

- (1) $2(\sqrt{2} + 2)$ (2) $2(\sqrt{2} + 1)$
(3) -5 (4) -7

Q128. Let $A = \begin{bmatrix} 2 & b & 1 \\ b & b^2 + 1 & b \\ 1 & b & 2 \end{bmatrix}$, where $b > 0$. Then the minimum value of $\frac{\det(A)}{b}$ is:

10 Jan 2019 (E)

- (1) $2\sqrt{3}$ (2) $-2\sqrt{3}$
(3) $\sqrt{3}$ (4) $-\sqrt{3}$

Q129. If the system of equations $x + y + z = 5$, $x + 2y + 3z = 9$, $x + 3y + \alpha z = \beta$ has infinitely many solutions, then $\beta - \alpha$ equals

10 Jan 2019 (M)

- (1) 8 (2) 21
(3) 5 (4) 18

Q130. The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$$\begin{aligned} x + 3y + 7z &= 0 \\ -x + 4y + 7z &= 0 \\ (\sin 3\theta)x + (\cos 2\theta)y + 2z &= 0 \end{aligned}$$

has a non-trivial solution, is:

10 Jan 2019 (E)

- (1) Two (2) Three
(3) Four (4) One

Q131. Let $a_1, a_2, a_3, \dots, a_{10}$ be in $G.P.$ with $a_i > 0$ for $i = 1, 2, \dots, 10$ and S be the set of pairs (r, k) , $r, k \in \mathbb{N}$ (the set of natural numbers) for which

$$\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$$

Then the number of elements in S , is:

10 Jan 2019 (E)

(1) Infinitely many

(2) 4

(3) 10

(4) 2

Q132. The system of linear equations

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

09 Jan 2019 (M)

(1) is inconsistent when $|a| = \sqrt{3}$

(2) has a unique solution for $|a| = \sqrt{3}$

(3) has infinitely many solutions for $a = 4$

(4) is inconsistent when $a = 4$

Q133. If the system of linear equations $x - 4y + 7z = g$; $3y - 5z = h$; $-2x + 5y - 9z = k$ is consistent, then:

09 Jan 2019 (E)

(1) $g + h + 2k = 0$

(2) $g + 2h + k = 0$

(3) $2g + h + k = 0$

(4) $g + h + k = 0$

ANSWER KEYS

1. (2)	2. (450)	3. (2)	4. (2)	5. (38)	6. (55)	7. (2)	8. (3)
9. (1)	10. (2)	11. (2)	12. (2)	13. (3)	14. (4)	15. (1)	16. (1)
17. (2)	18. (3)	19. (113)	20. (2)	21. (2)	22. (2)	23. (2)	24. (6)
25. (4)	26. (1)	27. (2)	28. (24)	29. (1)	30. (3)	31. (4)	32. (1)
33. (4)	34. (1)	35. (3)	36. (4)	37. (3)	38. (4)	39. (2)	40. (4)
41. (3)	42. (2)	43. (2)	44. (3)	45. (4)	46. (4)	47. (2)	48. (3)
49. (2)	50. (2)	51. (58)	52. (4)	53. (3)	54. (3)	55. (3)	56. (4)
57. (2)	58. (2)	59. (3)	60. (1)	61. (4)	62. (1)	63. (2)	64. (7)
65. (1)	66. (1)	67. (3)	68. (6)	69. (5)	70. (1)	71. (4)	72. (1)
73. (2)	74. (4)	75. (2)	76. (1)	77. (4)	78. (1)	79. (3)	80. (3)
81. (3)	82. (3)	83. (21)	84. (2)	85. (1)	86. (1)	87. (3)	88. (2)
89. (3)	90. (2)	91. (3)	92. (2)	93. (2)	94. (2)	95. (1)	96. (4)
97. (5)	98. (2)	99. (3)	100. (8)	101. (4)	102. (4)	103. (2)	104. (3)
105. (1)	106. (3)	107. (4)	108. (13)	109. (1)	110. (1)	111. (4)	112. (2)
113. (1)	114. (2)	115. (3)	116. (1)	117. (2)	118. (3)	119. (2)	120. (1)
121. (3)	122. (3)	123. (3)	124. (2)	125. (4)	126. (3)	127. (3)	128. (1)
129. (1)	130. (1)	131. (1)	132. (1)	133. (3)			

For solutions, [click here to download the MARKS App](#) from Google Play or visit web.getmarks.app on PC/desktop

